

Number Power

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

How Big Is a Million

Read the place, then say what the digit is worth. Type digits only — no commas needed.

Warm-Up 11 ITEMS • RECALL

1 How many tens in 100?

2 How many hundreds in 1,000?

3 10×100

4 100×100

5 $1,000 \times 10$

6 How many tens in 50

7 How many hundreds in 700

8 10×10

9 100×10

10 Zeros in ten thousand

11 How many tens in 1,000

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Write six million forty thousand five (6,040,005 — hold the empty places with zeros)

.....

13. The value of the 5 in 5,400

.....

14. The value of the 8 in 380

.....



Challenge NOT GRADED · REASON IT OUT

NOT GRADED • REASON IT OUT

15. How many thousands in 10,000?

16. In 3,472,861 — what is the 4 worth? (It sits in the hundred-thousands place)

17. In 3,472,861 — what is the 7 worth? (Ten-thousands place)

18. In 5,208,043 — what is the 2 worth? (Count the places from the right)

19. How many thousands make 4,000,000? ($4,000 \times 1,000 = 4,000,000$)

20. How many zeros in one million? (1,000,000)

21. $1,000 \times 1,000$ (A thousand thousands)

22. The value of the 2 in 24,000

23. Thousands in 250,000

24. Hundreds in 45,000

25. Millions in one billion

26. Zeros in a hundred million



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Ten Times Bigger

Multiplying by ten shifts every digit one place left. Watch the zeros.

✦ Warm-Up 9 ITEMS • RECALL

1 40×10

2 700×10

3 60×100

4 $8 \times 1,000$

5 $3,200 \div 10$

6 20×10

7 500×10

8 90×100

9 $4,000 \div 10$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. 230×100 (Two more zeros)

11. $9,000 \div 100$ (Take two zeros away)

12. 350×100

13. $1,200 \times 10$

14. $5,600 \div 10$



Challenge NOT GRADED · REASON IT OUT

NOT GRADED • REASON IT OUT

15. $45,000 \div 1,000$

16. $56 \times 1,000$ (Three more zeros)

17. $740,000 \div 10,000$ (Four zeros off)

18. How many times bigger is 800,000 than 800? (Three places left)

19. The 6 in 6,000 is worth how many times the 6 in 60? (Two places apart)

20. How many 10,000s make one million? $(1,000,000 \div 10,000)$

21. 7×100

22. $60,000 \div 1,000$

23. $84,000 \div 100$

24. $25 \times 1,000$

25. The 3 in 30,000 is worth how many times the 3 in 300

26. How many 1,000s make ten million



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Compare, Order, Round

Rounding depends on which place you were asked for. Read the question twice.

Warm-Up 12 ITEMS • RECALL

1 Round 47 to the nearest ten

2 Round 83 to the nearest ten

3 Round 65 to the nearest ten

4 Round 128 to the nearest ten

5 Round 254 to the nearest ten

6 Round 999 to the nearest ten

7 Round 34 to the nearest ten

8 Round 76 to the nearest ten

9 Round 55 to the nearest ten

10 Round 245 to the nearest hundred

11 Round 380 to the nearest hundred

12 Round 909 to the nearest ten

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 2,499 to the nearest hundred (Look at the tens digit only)

14. Round 3,472 to the nearest hundred

15. Round 6,509 to the nearest thousand



Challenge

NOT GRADED • REASON IT OUT

16. Round 47,382 to the nearest thousand (382 down beats 618 up)
17. Round 47,382 to the nearest ten thousand (Now the ends are 40,000 and 50,000)
18. Round 863,209 to the nearest hundred thousand (63,209 up to 900,000 is the shorter trip)
19. Larger number: 408,916 or 480,169? Type it. (Compare the ten-thousands place)
20. Largest number that rounds to 5,000 at the nearest thousand (One less than the halfway point)
21. Smallest number that rounds to 5,000 at the nearest thousand (Exactly halfway rounds up)
22. Round 24,650 to the nearest thousand
23. Round 89,999 to the nearest ten thousand
24. Round 12,345 to the nearest hundred
25. Largest whole number that rounds to 700 at the nearest hundred
26. Smallest whole number that rounds to 700 at the nearest hundred



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Order of Operations

Build the groups first, then add and subtract what's left loose.

Warm-Up 12 ITEMS • RECALL

1 $3 + 4 \times 2$

2 $10 - 2 \times 3$

3 $12 \div 4 + 5$

4 $6 + 6 \div 2$

5 $2 \times 3 + 4 \times 5$

6 $20 - (4 + 6)$

7 $5 + 2 \times 3$

8 $8 - 3 \times 2$

9 $16 \div 4 + 1$

10 $9 + 9 \div 3$

11 $(5 + 5) \times 2$

12 $30 - (5 + 5)$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $5 + 3 \times 4$ (Multiply first: $3 \times 4 = 12$)
.....

14. $(5 + 3) \times 4$ (Brackets overrule everything)
.....

15. $18 \div (3 + 6)$ ($3 + 6 = 9$ first)
.....

16. $40 - 6 \times 5 + 2$ ($40 - 30 + 2$)
.....

17. $3 \times (12 - 4) \div 6$ ($3 \times 8 = 24$, then $\div 6$)
.....

18. $4 \times 5 - 6 \div 2$
.....

19. $(12 + 8) \div 4$
.....

20. $7 \times (3 + 2)$
.....

21. $48 \div (2 \times 3)$
.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Powers & Expression Duel

Enrichment day. The little raised number counts copies, not multiples.



Working space



Error journal

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • How Big Is a Million

OUT OF 14

Warm-Up • 1-11 • 10 | 10 | 1000 | 10000 | 10000 | 5 | 7 | 100 | 1000 | 4 | 100

Core • 12-14 • 6040005 | 5000 | 80

Challenge • 15-26 • 10 | 400000 | 70000 | 200000 | 4000 | 6 | 1000000 | 20000 | 250 | 450 | 1000 | 8

1.2 • Ten Times Bigger

OUT OF 14

Warm-Up • 1-9 • 400 | 7000 | 6000 | 8000 | 320 | 200 | 5000 | 9000 | 400

Core • 10-14 • 23000 | 90 | 35000 | 12000 | 560

Challenge • 15-26 • 45 | 56000 | 74 | 1000 | 100 | 100 | 700 | 60 | 840 | 25000 | 100 | 10000

1.3 • Compare, Order, Round

OUT OF 15

Warm-Up • 1-12 • 50 | 80 | 70 | 130 | 250 | 1000 | 30 | 80 | 60 | 200 | 400 | 910

Core • 13-15 • 2500 | 3500 | 7000

Challenge • 16-26 • 47000 | 50000 | 900000 | 480169 | 5499 | 4500 | 25000 | 90000 | 12300 | 749 | 650

1.4 • Order of Operations

OUT OF 21

Warm-Up • 1-12 • 11 | 4 | 8 | 9 | 26 | 10 | 11 | 2 | 5 | 12 | 20 | 20

Core • 13-21 • 17 | 32 | 2 | 12 | 4 | 17 | 5 | 35 | 8

Challenge • 22-26 • 11 | 36 | 40 | 50 | 14

Fri • Powers & Expression Duel

ENRICHMENT

Challenge • 1-14 • 8 | 25 | 10000 | 25 | 24 | 81 | 1000 | 8 | 27 | 16 | 125 | 32 | 36 | 64

Rounding to Any Place

Find the place, look one to its right, decide.

Warm-Up 12 ITEMS • RECALL

1 Round 48 to the nearest ten

2 Round 43 to the nearest ten

3 Round 250 to the nearest hundred

4 Round 149 to the nearest hundred

5 Round 75 to the nearest ten

6 Round 612 to the nearest ten

7 Round 62 to the nearest ten

8 Round 67 to the nearest ten

9 Round 340 to the nearest hundred

10 Round 360 to the nearest hundred

11 Round 95 to the nearest ten

12 Round 450 to the nearest hundred

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 4,829 to the nearest hundred

14. Round 4,829 to the nearest thousand

15. Round 8,950 to the nearest hundred

16. Round 5,432 to the nearest thousand

17. Round 5,632 to the nearest thousand

18. Round 7,777 to the nearest hundred

Estimate Before You Compute

Round first. Then the real answer has something to be checked against.

Warm-Up 11 ITEMS • RECALL

1 Estimate $48 + 31$ by rounding both to tens

2 Estimate $62 - 19$

3 Estimate 19×21

4 Estimate $81 \div 9$

5 Estimate $297 + 104$

6 Estimate $52 + 48$

7 Estimate $39 + 42$ by rounding to tens

8 Estimate $71 - 28$

9 Estimate 29×11

10 Estimate $63 \div 8$

11 Estimate $198 + 205$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Estimate $4,812 + 3,190$ to the nearest thousand

.....

13. Estimate $612 - 388$ to the nearest hundred

.....

14. Estimate $3,964 \div 4$

.....

15. Estimate $486 + 317$ to the nearest hundred

.....

16. Estimate $4,120 - 1,890$ to the nearest hundred

.....

17. Estimate 396×21

.....

18. Estimate $5,940 \div 6$

.....

19. Estimate 78×41

.....

How Wrong Is the Estimate

An estimate you never compare to anything teaches you nothing.

Warm-Up 10 ITEMS • RECALL

1 Estimate of $48 + 31$

2 True value of $48 + 31$

3 The difference

4 True value of 19×21

5 Estimate of 19×21

6 Estimate of $39 + 42$

7 True value of $39 + 42$

8 True value of 29×11

9 Estimate of 29×11

10 The gap between them

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. True value of $612 - 388$

.....

12. Estimate of $612 - 388$ to the nearest hundred

.....

13. How far off was that estimate

.....

14. True value of $4,812 + 3,190$

.....

15. How far off was the thousand estimate

.....

16. Estimate of $486 + 317$

.....

17. True value of $486 + 317$

.....

18. The difference there

.....

19. True value of 78×41

.....

20. Estimate of 78×41

.....

Big Numbers in Context

Millions stop being abstract when you attach them to something real.

Warm-Up 11 ITEMS • RECALL

1 Zeros in one thousand

2 Zeros in one million

3 Thousands in a million

4 Hundreds in a thousand

5 Tens in a thousand

6 Zeros in ten thousand

7 Zeros in a hundred thousand

8 Ten thousands in a million

9 Hundreds in ten thousand

10 Tens in ten thousand

11 Thousands in ten thousand

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Days in about a million minutes, rounded to the nearest hundred (A million minutes is about 694 days)

13. Dots per page 1,000 — pages for a million dots

14. Seconds in an hour

15. Hours in a week

16. Minutes in a day

17. Days in ten years, ignoring leap days

18. Seconds in a day

A Million Dots, Measured

Prove the estimate. Count one square, scale it up, defend the number.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

 Warm-Up 4 ITEMS • RECALL

- 1 Dots in a 10 by 10 square

- 2 Dots in ten such squares

- 3 Dots in a 5 by 5 square

- 4 Dots in four such squares

 Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Squares of 100 dots needed for a million
.....
6. A 100 by 100 grid holds how many dots
.....
7. Dots in a 20 by 20 square
.....

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Rounding to Any Place

OUT OF 18

Warm-Up • 1-12 • 50 | 40 | 300 | 100 | 80 | 610 | 60 | 70 | 300 | 400 | 100 | 500

Core • 13-18 • 4800 | 5000 | 9000 | 5000 | 6000 | 7800

Challenge • 19-26 • 28000 | 100000 | 250 | 349 | 19000 | 46000 | 7500 | 8499

2.2 • Estimate Before You Compute

OUT OF 19

Warm-Up • 1-11 • 80 | 40 | 400 | 9 | 400 | 100 | 80 | 40 | 300 | 8 | 400

Core • 12-19 • 8000 | 200 | 1000 | 800 | 2200 | 8000 | 1000 | 3200

Challenge • 20-26 • 800 | 748 | 100 | 3072 | 500 | 240 | 3256

2.3 • How Wrong Is the Estimate

OUT OF 20

Warm-Up • 1-10 • 80 | 79 | 1 | 399 | 400 | 80 | 81 | 319 | 300 | 19

Core • 11-20 • 224 | 200 | 24 | 8002 | 2 | 800 | 803 | 3 | 3198 | 3200

Challenge • 21-24 • high | 1599 | low | 2499

2.4 • Big Numbers in Context

OUT OF 18

Warm-Up • 1-11 • 3 | 6 | 1000 | 10 | 100 | 4 | 5 | 100 | 100 | 1000 | 10

Core • 12-18 • 700 | 1000 | 3600 | 168 | 1440 | 3650 | 86400

Challenge • 19-22 • 3000 | 100 | 99 | 12

Fri • A Million Dots, Measured

OUT OF 8

Warm-Up • 1-4 • 100 | 1000 | 25 | 100

Core • 5-8 • 10000 | 10000 | 400 | 100

Challenge • 9-14 • 100 | 20 | 1000000 | 500 | 50000 | 100

Why Order Matters

Without an agreed order, one expression has several answers.

Warm-Up 12 ITEMS • RECALL

1 $2 + 3 \times 4$

2 $(2 + 3) \times 4$

3 $10 - 2 \times 3$

4 $(10 - 2) \times 3$

5 $6 + 6 \div 2$

6 $(6 + 6) \div 2$

7 $3 + 2 \times 5$

8 $(3 + 2) \times 5$

9 $12 - 3 \times 2$

10 $(12 - 3) \times 2$

11 $8 + 8 \div 4$

12 $(8 + 8) \div 4$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $5 + 4 \times 3 - 2$

14. $(5 + 4) \times (3 - 2)$

15. $20 - 3 \times 4 + 2$

16. $36 \div (2 + 4)$

17. $36 \div 2 + 4$

18. $20 - 3 \times 4 + 1$

19. $(20 - 3) \times 4$

20. $2 \times 6 + 3 \times 4$

21. $2 \times (6 + 3) \times 4$

22. $36 \div 6 \div 2$

Brackets First

Whatever is inside the brackets happens before anything else.

Warm-Up 12 ITEMS • RECALL

1 $(4 + 5) \times 2$

2 $(9 - 3) \div 2$

3 $3 \times (2 + 2)$

4 $(7 + 3) \times 10$

5 $(12 - 4) \div 4$

6 $5 \times (10 - 6)$

7 $(2 + 6) \times 3$

8 $(15 - 5) \div 5$

9 $4 \times (3 + 3)$

10 $(9 + 1) \times 10$

11 $(20 - 8) \div 4$

12 $6 \times (12 - 8)$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $(15 + 5) \div (2 + 2)$

14. $4 \times (6 + 3) - 10$

15. $(8 \times 3) - (4 \times 2)$

16. $2 \times (50 - 15)$

17. $(25 - 9) \div (2 \times 4)$

18. $((3 + 3) \times 4) - 10$

19. $100 \div (5 \times 5)$

20. $(7 + 8) \times (10 - 6)$

21. $(45 \div 9) + (6 \times 3)$

★ Challenge NOT GRADED • REASON IT OUT

22. $100 - (25 \times 3)$

23. $((4 + 2) \times 3) - 8$

24. $120 \div (2 \times (3 + 3))$

25. $((10 + 2) \times 5) \div 3$

26. $240 \div (4 \times (2 + 1))$

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Multi-Step Expressions

Multiply and divide before you add and subtract. Left to right within each.

✦ Warm-Up 12 ITEMS • RECALL

1 $12 \div 3 + 2$

2 $12 + 3 \times 2$

3 $20 \div 4 \times 2$

4 $18 - 6 \div 3$

5 $7 \times 2 + 6$

6 $30 \div 5 - 2$

7 $18 \div 3 + 4$

8 $18 + 3 \times 4$

9 $24 \div 6 \times 2$

10 $20 - 8 \div 4$

11 $9 \times 3 + 3$

12 $40 \div 8 - 3$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $48 \div 6 + 3 \times 4$

14. $7 \times 8 - 6 \times 5$

15. $64 \div 8 \div 2$

16. $9 + 36 \div 4 - 3$

17. $60 \div 5 + 7 \times 2$

18. $81 \div 9 - 3 \times 2$

19. $7 \times 8 + 100 \div 4$

★

Challenge

NOT GRADED · REASON IT OUT

20. $100 - 4 \times 12$

.....

21. $144 \div 12 + 8 \times 3 - 5$

.....

22. $2 \times 3 + 4 \times 5 + 6 \times 7$

.....

23. $11 \times 4 - 22 \div 2$

.....

24. $144 \div 12 \times 3$

.....

25. $200 \div 8 + 6 \times 7 - 9$

.....

26. $3 \times 4 + 5 \times 6 + 7 \times 8$

.....

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Write the Expression

Turn the sentence into symbols, then work it out.

Warm-Up 12 ITEMS • RECALL

1 Five more than three lots of four

2 Double seven, then add one

3 Ten less than six times three

4 Half of twenty, plus four

5 Three lots of nine

6 Twenty shared by four

7 Four more than double six

8 Triple five, then add two

9 Twenty less than five times six

10 A third of thirty, plus one

11 Two lots of nine, minus eight

12 Half of sixteen, times three

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. The sum of 8 and 4, multiplied by 3

.....

14. 8 added to 4 lots of 3

.....

15. 100 less the product of 7 and 9

.....

16. The difference of 20 and 8, divided by 4

.....

17. Six lots of the sum of 2 and 3

.....

18. Six times the sum of two and five

.....

19. The product of 7 and 8, less 20

.....

20. Divide 96 by the sum of 4 and 8

.....

21. Add 15 to the product of 6 and 9

.....

22. Double the difference between 30 and 12

.....



Working space



Error journal

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 Round 4,829 to the nearest hundred

2 $2 + 3 \times 4$

3 Round 3,451 to the nearest hundred

4 $5 + 2 \times 3$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $(5 + 4) \times (3 - 2)$

6. $48 \div 6 + 3 \times 4$

7. Zeros in one million

8. The sum of 8 and 4, multiplied by 3

9. $(20 - 3) \times 4$

10. Estimate 396×21

 Challenge NOT GRADED · REASON IT OUT

11. Estimate 187×4

12. $144 \div 12 + 8 \times 3 - 5$

13. $5 + 4 \times 3 - 12 \div 4$

14. Largest whole number that rounds to 700 at the nearest hundred

- ### 15. Seconds in a day



Working space



Error journal

Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

3.1 • Why Order Matters

OUT OF 22

Warm-Up • 1-12 • 14 | 20 | 4 | 24 | 9 | 6 | 13 | 25 | 6 | 18 | 10 | 4

Core • 13-22 • 15 | 9 | 10 | 6 | 22 | 9 | 68 | 24 | 72 | 3

Challenge • 23-26 • 70 | 50 | 54 | 30

3.2 • Brackets First

OUT OF 21

Warm-Up • 1-12 • 18 | 3 | 12 | 100 | 2 | 20 | 24 | 2 | 24 | 100 | 3 | 24

Core • 13-21 • 5 | 26 | 16 | 70 | 2 | 14 | 4 | 60 | 23

Challenge • 22-26 • 25 | 10 | 10 | 20 | 20

3.3 • Multi-Step Expressions

OUT OF 19

Warm-Up • 1-12 • 6 | 18 | 10 | 16 | 20 | 4 | 10 | 30 | 8 | 18 | 30 | 2

Core • 13-19 • 20 | 26 | 4 | 15 | 26 | 3 | 81

Challenge • 20-26 • 52 | 31 | 68 | 33 | 36 | 58 | 98

3.4 • Write the Expression

OUT OF 22

Warm-Up • 1-12 • 17 | 15 | 8 | 14 | 27 | 5 | 16 | 17 | 10 | 11 | 10 | 24

Core • 13-22 • 36 | 20 | 37 | 3 | 30 | 42 | 36 | 8 | 69 | 36

Challenge • 23-26 • 24 | 4 | 48 | 10

Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 4800 | 14 | 3500 | 11

Core • 5-11 • 9 | 20 | 800 | 6 | 36 | 68 | 8000

Challenge • 12-15 • 31 | 14 | 749 | 86400

The Little Raised Number

An exponent counts how many times the base is multiplied by itself.

✦ Warm-Up 10 ITEMS • RECALL

1

 2^2

2

 3^2

3

 5^2

4

 2^3

5

 4^2

6

 6^2

7

 7^2

8

 8^2

9

 9^2

10

 1^5

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 3^3

.....

12. 2^4

.....

13. 5^3

.....

14. 2^5

.....

15. 4^3

.....

16. 11^2

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Powers of Ten

The exponent is the number of zeros. That is the whole shortcut.

Warm-Up 3 ITEMS • RECALL

1 10^1 2 Zeros in 10^7 3 10^0

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

4. Write 5,000 as a digit times a power of ten — type the exponent

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Expanded Form with Powers

Every place value is a power of ten in disguise.

Warm-Up 12 ITEMS • RECALL

1 The value of the 4 in 400

2 The value of the 7 in 70

3 The value of the 3 in 3,000

4 $200 + 30 + 5$

5 $4,000 + 200 + 10$

6 The value of the 9 in 900

7 How many hundreds make 2,600

8 How many hundreds make 4,300

9 How many hundreds make 7,100

10 How many hundreds make 9,500

11 How many tens make 3,000

12 How many tens make 6,000

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

20. In 4,271 — what is the 4 worth

.....

21. In 4,271 — what is the 2 worth

.....

22. In 4,271 — what is the 7 worth

.....

23. In 8,605 — what is the 8 worth

.....

24. In 8,605 — what is the 6 worth

.....

25. In 3,940 — what is the 9 worth

.....

26. In 7,128 — what is the 7 worth

.....

27. In 7,128 — what is the 1 worth

.....

28. In 5,083 — what is the 5 worth

.....

29. Bigger: 3,421 or 3,412

.....

30. Bigger: 5,680 or 5,806

.....



Challenge NOT GRADED · REASON IT OUT

NOT GRADED • REASON IT OUT

33. $6 \times 10^3 + 4 \times 10^2 + 2$

34. $3 \times 10^4 + 5 \times 10^2$

35. The value of the 8 in 84,000

36. $9 \times 10^2 + 9 \times 10^1 + 9$

37. 7×10^5

38. $2 \times 10^5 + 6 \times 10^3 + 4 \times 10^1$

39. The largest power of ten inside 47,300 — type its exponent

40. $5 \times 10^3 + 2 \times 10^2$

41. $7 \times 10^4 + 3 \times 10^2$

42. $4 \times 10^3 + 6 \times 10^1 + 2$

43. $20,000 + 4,000 + 300 + 50$

44. $3 \times 10^5 + 7 \times 10^3 + 5 \times 10^1$



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

Warm-Up 10 ITEMS • RECALL

1 Round 149 to the nearest hundred

2 $2 + 3 \times 4$

3 Zeros in one million

4 5^2

5 $(2 + 3) \times 4$

6 Round 340 to the nearest hundred

7 $5 + 2 \times 3$

8 Zeros in ten thousand

9 6^2

10 $(3 + 2) \times 5$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $48 \div 6 + 3 \times 4$

.....

12. Estimate $3,964 \div 4$

.....

13. Estimate $486 + 317$ to the nearest hundred

.....

14. $60 \div 5 + 7 \times 2$

.....

15. 2^4

.....



Challenge

16. 10^3

17. Round 96,412 to the nearest ten thousand

18. 4×10^4

19. $6 \times 10^3 + 4 \times 10^2 + 2$

20. $144 \div 12 + 8 \times 3 - 5$

21. 2^8

22. 10^4

23. Round 24,650 to the nearest thousand

24. $4 \times 10^3 + 6 \times 10^1 + 2$

25. $200 \div 8 + 6 \times 7 - 9$

26. 2^{10}



Working space



Error journal

Mission 03 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 Round 612 to the nearest ten
_____2 3^2
_____3 Round 448 to the nearest ten
_____4 4^2

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.5. Round 4,829 to the nearest thousand
.....6. $10 - 2 \times 3$
.....7. $(15 + 5) \div (2 + 2)$
.....8. 2^5
.....9. Estimate $4,812 + 3,190$ to the nearest thousand
.....10. The sum of 12 and 8, divided by the difference of 9 and 4
.....11. $(15 - 5) \div 5$
.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • The Little Raised Number

OUT OF 16

Warm-Up • 1-10 • 4 | 9 | 25 | 8 | 16 | 36 | 49 | 64 | 81 | 1

Core • 11-16 • 27 | 16 | 125 | 32 | 64 | 121

Challenge • 17-22 • 100 | 1000 | 256 | 81 | 1024 | 64

4.2 • Powers of Ten

OUT OF 4

Warm-Up • 1-3 • 10 | 7 | 1

Core • 4 • 3

Challenge • 5-16 • 100 | 1000 | 10000 | 5 | 1000000 | 300 | 7000 | 40000 | 2500 |
100000 | 6 | 100000

4.3 • Expanded Form with Powers

OUT OF 23

Warm-Up • 1-12 • 400 | 70 | 3000 | 235 | 4210 | 900 | 26 | 43 | 71 | 95 | 300 |
600

Core • 20-30 • 4000 | 200 | 70 | 8000 | 600 | 900 | 7000 | 100 | 5000 | 3,421 |
5,806

Challenge • 33-44 • 6402 | 30500 | 80000 | 999 | 700000 | 206040 | 4 | 5200 | 70300 |
4062 | 24350 | 307050

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 100 | 14 | 1000 | 6 | 25 | 20 | 300 | 11 | 10000 | 4 | 36 | 25

Core • 13-22 • 100000 | 20 | 40000 | 1000 | 6402 | 800 | 26 | 25000 | 4062 | 16

Challenge • 23-26 • 31 | 256 | 58 | 1024

Fri • Mission 03 Test

OUT OF 14

Warm-Up • 1-4 • 610 | 9 | 450 | 16

Core • 5-14 • 5000 | 4 | 5 | 52 | 7000 | 32 | 8000 | 4 | 2 | 70000

Challenge • 15-19 • 206040 | 68 | 98 | 12 | 125